

Problem-Solution Fit

Date	28 June 2026
Team ID	SWTID-2026-9097
Project Name	Voice--Based--Concept--Understanding--Analyzer
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

Canvas Section	Details
1. Customer Segment(s) [CS]	• Students (School & College) • Teachers & Faculty • Interview candidates • Online learners and training institutes
2. Problems / Pains + Frequency [PR]	• Difficulty evaluating conceptual understanding through spoken explanations (Very Frequent). • Manual evaluation consumes significant time (Frequent). • Lack of objective and consistent feedback (Frequent). • Students do not know where they need improvement (Frequent).
3. Triggers to Act [TR]	• Preparing for exams or viva voce. • Need for quick concept assessment. • Requirement for self-learning and practice. • Faculty requiring automated evaluation.
4. Emotions Before / After [EM]	Before: Nervous, uncertain, anxious, lacking confidence. After: Confident, motivated, satisfied, aware of strengths and weaknesses.
5. Available Solutions – Pros & Cons [AS]	Manual Teacher Evaluation: Accurate but time-consuming. Online Quiz Platforms: Fast but only assess written answers. Speech-to-Text Tools: Convert speech but do not evaluate understanding. AI Chatbots: Provide explanations but lack structured voice assessment and scoring.
6. Customer Limitations [CL]	• Limited internet connectivity. • Low-end devices. • Background noise during recording. • Limited technical knowledge. • Budget constraints for premium educational tools.
7. Behavior + Intensity [BE]	• Frequently practice explanations before exams. • Record voice multiple times to improve. • Seek immediate feedback after practice. Intensity: High during examinations and interviews.
8. Channels of Behavior [CH]	Online: Educational websites, LMS platforms, mobile applications, university portals, online courses. Offline: Classroom presentations, viva examinations, seminars, training sessions, workshops.
9. Problem Root / Cause [RC]	• Lack of automated tools for evaluating spoken conceptual understanding. • Dependence on manual evaluation. • Inconsistent assessment methods. • No personalized feedback for improvement.

Canvas Section	Details
10. Your Solution [SL]	Voice-Based Concept Understanding Analyser (VBCUA) enables users to upload or record voice explanations. The system uses Whisper for speech-to-text, Sentence-BERT for semantic similarity, Librosa for speech feature extraction, and Gemini AI for intelligent feedback. It generates performance scores, visual analysis, personalized recommendations, and downloadable PDF reports, providing fast, objective, and accurate concept evaluation.